

Model Selection and Smoothing in Generalized Nonparametric Regression

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Multiple and Generalized Nonparametric Regression

Suppose that Y that follows an exponential family distribution.

Given predictors $(X_1, \dots, X_p)$, consider a MGNR model of the form

$$g(\mu) = f(X_1, \dots, X_p)$$

where

- μ is the (conditional) expectation of Y (given $X_1, \dots, X_p$)
- $g(\cdot)$ is a user-specified link function (monotonic and invertible)
- $f(\cdot)$ is an unknown “smooth” function of $q \leq p$ predictors

Goal: estimate the unknown function $f(\cdot)$ from training data.

Challenge: Model Structure is Completely Unknown

Variable Selection Problem: which predictors should be included?

- Not all of the candidate predictors may be needed
- Want to determine which predictors are relevant (active)

Term Selection Problem: how should variables be included?

- Do the predictors combine in an additive fashion? Or interactive?
- Want to learn which main and/or interaction effects matter

Smoothing Problem: how smooth should each term be?

- Predictors may (and likely will) enter model in nonlinear fashion
- Want to avoid under-fitting (bias) and over-fitting (variability)

Three-Pronged Approach

Step 1: The Algorithm (Adaptively Bounded Fisher Scoring)

- Existing algorithms aren't easily adaptable (e.g., to different g)
- Develop versatile group elastic net algorithm for GLMs

Step 2: The Theory (Tensor Product Reproducing Kernels)

- Existing tensor product RKHS theory isn't ideal for selection
- Develop improved representer theorem for tensor products

Step 3: The Practice (Smoothing and Subspace Selection)

- Existing implementations only consider additive models
- Develop intuitive software for selecting main and interaction effects

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Generalized Linear Model

Consider a generalized linear model (GLM) of the form

$$g(\mu) = \mathbf{X}^\top \boldsymbol{\beta} \quad (1)$$

where

- $g(\cdot)$ is an invertible link function
- $\mu = E(Y|\mathbf{X})$ is the conditional expectation of Y given $\mathbf{X}$
- $\mathbf{X} = (X_1, \dots, X_p)^\top \in \mathbb{R}^p$ is the predictor vector
- $\boldsymbol{\beta} \in \mathbb{R}^p$ is the unknown coefficient vector

The maximum likelihood (ML) estimates of the coefficients are

$$\hat{\boldsymbol{\beta}} = \arg \min_{\boldsymbol{\beta} \in \mathbb{R}^p} L(\boldsymbol{\beta}|\mathbf{D})$$

where

- $L(\boldsymbol{\beta}|\mathbf{D})$ denotes the negative of the log-likelihood function
- $\mathbf{D} = \{\mathbf{y}, \mathbf{X}\}$ is the observed data

Ridge, Lasso, and Elastic Net

Consider finding the coefficients that satisfy

$$\hat{\boldsymbol{\beta}}_{\lambda, \alpha}^* = \arg \min_{\boldsymbol{\beta} \in \mathbb{R}^p} \left\{ \frac{1}{n} L(\boldsymbol{\beta} | \mathbf{D}) + \lambda P_{\alpha}^*(\boldsymbol{\beta}) \right\}$$

where $\lambda \geq 0$ controls the overall influence of the penalty

$$P_{\alpha}^*(\boldsymbol{\beta}) = \alpha \sum_{j=1}^p |\beta_j| + \frac{1}{2}(1 - \alpha) \sum_{j=1}^p \beta_j^2 \quad (2)$$

and $\alpha \in [0, 1]$ controls the weight of the lasso and ridge penalties.

- $\alpha = 0$ corresponds to the ridge (Hoerl & Kennard, 1970)
- $\alpha = 1$ corresponds to the lasso (Tibshirani, 1996)
- $\alpha \in (0, 1)$ corresponds to the elastic net (Zou & Hastie, 2005)

Regularization with Grouped Predictors

Multiple β_j s may be needed to represent a single model term:

- categorical predictors, polynomial effects, regression splines
- interactions involving any of the above

Problem: the penalty in Equation (2) selects β_j s individually (if $\alpha > 0$).

Solution(ish): Yuan & Lin (2006) proposed the group lasso penalty

$$P(\boldsymbol{\beta}) = \sum_{k=1}^K \omega_k \|\boldsymbol{\beta}_k\| \quad (3)$$

where

- $\omega_k \geq 0$ weights the k -th group's penalty (typically $\omega_k = \sqrt{p_k}$)
- $\|\boldsymbol{\beta}_k\| = (\boldsymbol{\beta}_k^\top \boldsymbol{\beta}_k)^{1/2}$ is the Euclidean norm of $\boldsymbol{\beta}_k$

Group Elastic Net Regularized GLM

I consider a groupwise extension of the classic elastic net penalty:

$$\hat{\boldsymbol{\beta}}_{\lambda,\alpha} = \arg \min_{\boldsymbol{\beta} \in \mathbb{R}^p} \left\{ \frac{1}{n} L(\boldsymbol{\beta} | \mathbf{D}) + \lambda P_{\alpha}(\boldsymbol{\beta}) \right\} \quad (4)$$

where the group elastic net penalty function has the form

$$P_{\alpha}(\boldsymbol{\beta}) = \sum_{k=1}^K \omega_k \left(\alpha \|\boldsymbol{\beta}_k\| + \frac{1}{2} (1 - \alpha) \|\boldsymbol{\beta}_k\|^2 \right) \quad (5)$$

which is a natural groupwise extension of Equation (2).

Exponential Family Form

The response variable Y follows an exponential family distribution:

$$f_Y(y|\theta, \phi) = \exp \left\{ \frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi) \right\}$$

where

- θ is the canonical parameter
- ϕ is the dispersion parameter
- $a(\cdot)$, $b(\cdot)$, and $c(\cdot)$ are known functions

It is well-known that

$$E(Y) = \mu = b'(\theta)$$

$$\text{Var}(Y) = \sigma^2 = b''(\theta)a(\phi)$$

where

- $b'(\cdot) = db(\theta)/d\theta$ and $b''(\cdot) = d^2b(\theta)/d\theta^2$
- $b''(\theta) = V(\theta)$ is referred to as the variance function

Likelihood Function

Given an independent sample of n observations, the log-likelihood function has the form

$$l(\boldsymbol{\beta}|\mathbf{D}) = \sum_{i=1}^n \left(\frac{y_i \theta_i - b(\theta_i)}{\phi/w_i} + c(y_i, \phi) \right)$$

where

- y_i is the i -th observation's response
- $\theta_i = \tilde{g}(g^{-1}(\eta_i))$ is the i -th observation's canonical parameter
- $\tilde{g}(\cdot)$ is the canonical link and $g(\cdot)$ is the chosen link
- $\eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}$ is the i -th observation's linear predictor
- $\mathbf{x}_i = (x_{i1}, \dots, x_{ip})^\top$ is the i -th observation's predictor vector
- $w_i > 0$ is the i -th observation's known weight

Note: this assumes that $a(\phi) = \phi/w$ for some known $w > 0$.

Penalized Likelihood Estimation

Dropping $c(y_i, \phi)$ and multiplying by $-\frac{1}{n}$ produces the loss function

$$L(\boldsymbol{\beta}|\mathbf{D}) = -\frac{1}{n} \sum_{i=1}^n w_i (y_i \theta_i - b(\theta_i))$$

which sets $\phi = 1$ (WLOG) for the purposes of estimating $\boldsymbol{\beta}$.

The desired penalized likelihood estimates satisfy

$$\hat{\boldsymbol{\beta}}_{\lambda, \alpha} = \arg \min_{\boldsymbol{\beta} \in \mathbb{R}^p} \left\{ -\frac{1}{n} \sum_{i=1}^n w_i (y_i \theta_i - b(\theta_i)) + \lambda P_{\alpha}(\boldsymbol{\beta}) \right\}$$

where $\theta_i = \tilde{g}(g^{-1}(\mathbf{x}_i^{\top} \boldsymbol{\beta}))$ implicitly depends on the coefficient vector $\boldsymbol{\beta}$

Groupwise Penalized Likelihood Estimation

Focus on cyclical algorithms that iteratively estimate β_k given others.

I consider iteratively solving (for $k = 1, \dots, K$) the problem

$$\hat{\beta}_{\lambda, \alpha}^{(k)} = \arg \min_{\beta_k \in \mathbb{R}^{p_k}} \left\{ L(\beta | \mathbf{D}) + \lambda P_{\alpha}^{(k)}(\beta_k) \right\} \quad (6)$$

where

$$P_{\alpha}^{(k)}(\beta_k) = \omega_k \left(\alpha \|\beta_k\| + \frac{1}{2} (1 - \alpha) \|\beta_k\|^2 \right)$$

is the k -th group's penalty contribution.

Problem: there doesn't exist a closed-form solution for general cases.

Gradient Vector

The negative gradient of $L(\boldsymbol{\beta}|\mathbf{D})$ with respect to $\boldsymbol{\beta}_k$ is

$$\mathbf{g}_k = - \left(\frac{\partial L(\boldsymbol{\beta}|\mathbf{D})}{\partial \boldsymbol{\beta}_k} \right) = \frac{1}{n} \sum_{i=1}^n \frac{w_i}{d_i v_i} (y_i - \mu_i) \mathbf{x}_{ik} \quad (7)$$

where

- $v_i = b''(\theta_i)$ is the i -th observation's variance function realization
- $d_i = \left(\frac{d\eta}{d\mu} \Big|_{\mu=\mu_i} \right)$ is the derivative $g'(\mu)$ evaluated at $\mu_i = g^{-1}(\mathbf{x}_i^\top \boldsymbol{\beta})$

Table 1 gives the v_i for many commonly used response distributions.
Table 2 gives the d_i for many commonly used link functions.

Variance Functions

Table 1: Exponential family parameterizations and variance functions.

Family	Parameterization	ϕ	v_i
Gaussian	$N(\mu, \sigma^2)$	σ^2	1
Binomial	$B(\mu)$	1	$\mu_i(1 - \mu_i)$
Multinomial	$M(\mu)$	1	$\mu_{il}(1 - \mu_{il})$
Poisson	$P(\mu)$	1	μ_i
Negative Binomial	$NB(\mu, \theta)$	1	$\mu_i + \mu_i^2/\theta$
Gamma	$G(\mu, \nu)$	$1/\nu$	μ_i^2
Inverse Gaussian	$IG(\mu, \sigma^2)$	σ^2	μ_i^3

Link Functions

Table 2: Link functions $g(\cdot)$, their inverses $g^{-1}(\cdot)$, and their derivatives $g'(\cdot)$.

Link	$\eta = g(\mu)$	$\mu = g^{-1}(\eta)$	$d_i = g'(\mu_i)$
Identity	μ	η	1
Logit	$\log(\frac{\mu}{1-\mu})$	$\frac{\exp(\eta)}{1+\exp(\eta)}$	$\frac{1}{\mu_i(1-\mu_i)}$
Multi-Logit	N/A	$\frac{\exp(\eta_\ell)}{\sum_{l=1}^m \exp(\eta_l)}$	$\frac{1}{\mu_{i\ell}(1-\mu_{i\ell})}$
Log	$\log(\mu)$	$\exp(\eta)$	$1/\mu_i$
Inverse	$1/\mu$	$1/\eta$	$-1/\mu_i^2$
Inverse-Square	$1/\mu^2$	$1/\eta^{1/2}$	$-2/\mu_i^3$

Penalty Derivatives

Assuming $\beta_k \neq \mathbf{0}_{p_k}$, the gradient of the group lasso penalty $P_1(\beta)$ is

$$\mathbf{q}_{1k} = \frac{\partial P_1(\beta)}{\partial \beta_k} = \frac{\omega_k}{\|\beta_k\|} \beta_k$$

and note that $\mathbf{q}_{1k}$ is not defined if $\beta_k = \mathbf{0}_{p_k}$.

The gradient of the group ridge penalty $P_2(\beta)$ is

$$\mathbf{q}_{2k} = \frac{\partial P_2(\beta)}{\partial \beta_k^\top} = \omega_k \beta_k$$

for all $\beta_k \in \mathbb{R}^{p_k}$.

Optimality Conditions

A necessary condition for the vector $\boldsymbol{\beta}^\top = (\boldsymbol{\beta}_1^\top, \dots, \boldsymbol{\beta}_K^\top)$ to be a solution to Equation (4) is that

$$\begin{aligned} -\mathbf{g}_k + \lambda(\alpha \mathbf{q}_{1k} + (1 - \alpha) \mathbf{q}_{2k}) &= \mathbf{0}_{p_k} & \forall \boldsymbol{\beta}_k \neq \mathbf{0}_{p_k} \\ \|\mathbf{g}_k\| &\leq \alpha \lambda \omega_k & \forall \boldsymbol{\beta}_k = \mathbf{0}_{p_k} \end{aligned} \quad (8)$$

which follows directly from the Karush-Kuhn-Tucker (KKT) conditions (Karush, 1939; Kuhn & Tucker, 1951; Yang & Zou, 2015).

Fisher Information and Unpenalized Solution

The Fisher information matrix for the loss function L has the form

$$\mathbf{H}_k = E \left(\frac{\partial^2 L(\boldsymbol{\beta} | \mathbf{D})}{\partial \boldsymbol{\beta}_k \partial \boldsymbol{\beta}_k^\top} \right) = \frac{1}{n} \sum_{i=1}^n \frac{w_i}{d_i^2 v_i} \mathbf{x}_{ik} \mathbf{x}_{ik}^\top \quad (9)$$

which is positive definite if columns of $\mathbf{X}_k$ are linearly independent.

When $\lambda = 0$, the MLEs can be obtained via cyclic Fisher scoring:

$$\boldsymbol{\beta}_k^{(t+1)} = \boldsymbol{\beta}_k^{(t)} + \left(\mathbf{H}_k^{(t)} \right)^{-1} \mathbf{g}_k^{(t)} \quad (10)$$

where

- $\mathbf{g}_k^{(t)}$ and $\mathbf{H}_k^{(t)}$ are gradient and information evaluated at $\boldsymbol{\beta}_k^{(t)}$
- $t = 0, 1, 2, \dots$ denotes the iteration (step) counter

Local Quadratic Approximation

$$\boldsymbol{\beta}_k^* = \left(\tilde{\boldsymbol{\beta}}_1^\top, \dots, \tilde{\boldsymbol{\beta}}_K^\top \right) \text{ with } \tilde{\boldsymbol{\beta}}_j = \boldsymbol{\beta}_j \text{ for } j \in \{1, \dots, K\} \setminus \{k\} \text{ and } \tilde{\boldsymbol{\beta}}_k \neq \boldsymbol{\beta}_k.$$

Local quadratic approximation of $L(\boldsymbol{\beta}|\mathbf{D})$ near $\boldsymbol{\beta}_k^*$ has the form

$$\begin{aligned} L(\boldsymbol{\beta}|\mathbf{D}) &\approx L_k(\boldsymbol{\beta}_k|\mathbf{D}, \boldsymbol{\beta}_k^*) \\ &= L(\boldsymbol{\beta}_k^*|\mathbf{D}) - \left(\boldsymbol{\beta}_k - \tilde{\boldsymbol{\beta}}_k \right)^\top \tilde{\mathbf{g}}_k + \frac{1}{2} \left(\boldsymbol{\beta}_k - \tilde{\boldsymbol{\beta}}_k \right)^\top \tilde{\mathbf{H}}_k \left(\boldsymbol{\beta}_k - \tilde{\boldsymbol{\beta}}_k \right) \end{aligned}$$

where $\tilde{\mathbf{g}}_k$ and $\tilde{\mathbf{H}}_k$ denote gradient and information evaluated at $\tilde{\boldsymbol{\beta}}_k$.

Setting the derivative to zero to solve for $\boldsymbol{\beta}_k$ gives

$$\boldsymbol{\beta}_k = \tilde{\boldsymbol{\beta}}_k + \tilde{\mathbf{H}}_k^{-1} \tilde{\mathbf{g}}_k$$

which is Equation (10) with $\tilde{\boldsymbol{\beta}}_k$ serving as the current estimate

Modified Loss Function and Unpenalized Update

Consider minimizing the modified loss function

$$L_k^*(\beta_k | \mathbf{D}, \beta_k^*) = L(\beta_k^* | \mathbf{D}) - (\beta_k - \tilde{\beta}_k)^\top \tilde{\mathbf{g}}_k + \frac{\gamma_k \delta}{2} \|\beta_k - \tilde{\beta}_k\|^2$$

where

- γ_k is the largest eigenvalue of $\frac{1}{n} \mathbf{X}_k^\top \mathbf{W} \mathbf{X}_k$
- $\delta = \max_i (1/d_i^2 v_i) = \min_i (d_i^2 v_i)$ (fixed or adaptively updated)

Lemma: L_k^* majorizes the quadratic approximation function L_k , i.e., $L_k(\beta_k | \mathbf{D}, \beta_k^*) \leq L_k^*(\beta_k | \mathbf{D}, \beta_k^*)$ for all $\beta_k \in \mathbb{R}^{p_k}$.

The modified (majorized) Fisher scoring update

$$\beta_k^{(t+1)} = \beta_k^{(t)} + (\gamma_k \delta_{(t)})^{-1} \mathbf{g}_k^{(t)} \quad (11)$$

where $\delta_{(t)}$ is defined using the current coefficient estimates $\beta_k^{(t)}$.

Adaptive Gradient Descent Algorithm

Consider finding the β_k that satisfies

$$\arg \min_{\beta_k \in \mathbb{R}^{p_k}} \left\{ L_k^*(\beta_k | \mathbf{D}, \beta_k^*) + \lambda P_\alpha^{(k)}(\beta_k) \right\} \quad (12)$$

which is Equation (6) with L_k^* replacing L_k .

Theorem: The solution to Equation (6) can be obtained by cyclically solving Equation (12) using updates of the form

$$\beta_k^{(t+1)} = \frac{1}{1 + (1 - \alpha)\lambda\omega_k(\gamma_k\delta_{(t)})^{-1}} \left(1 - \frac{\alpha\lambda\omega_k(\gamma_k\delta_{(t)})^{-1}}{\|\mathbf{b}_k^{(t+1)}\|} \right) \mathbf{b}_k^{(t+1)} +$$

where

- $\mathbf{b}_k^{(t+1)} = \beta_k^{(t)} + (\gamma_k\delta_{(t)})^{-1} \mathbf{g}_k^{(t)}$ is the unpenalized update
- $\delta_{(t)} = \min_i (d_i^2 v_i)$ where d_i and v_i are evaluated using $\beta_k^{(t)}$

Relation to Existing Algorithms

Closely related to the update proposed by Yang & Zou (2015):

- $\alpha = 1$ and $\delta_{(t)} = \delta_* \forall t$ (i.e., no separation of γ_k and δ)
- Only consider Gaussian, binomial, and two SVM loss functions

Also related to the update proposed by Breheny & Huang (2015):

- $\alpha = 1$, $\gamma_k = 1$ (orthogonalized), and $\delta_{(t)} = \delta_* \forall t$
- Only consider Gaussian, binomial, Poisson, and Cox PH

Neither separates $\mathbf{H}_k$ into fixed design contribution (bounded by γ_k) and adaptive mean-dependent contribution (bounded by δ).

(Non-Exhaustive?) List of R Packages for Group Lasso

There are a handful of options for fitting group lasso in R:

- **gglasso** package (Yang et al., 2020)
- **grplasso** package (Meier, 2020)
- **grpnet** package (Helwig, 2023)
- **grpreg** package (Breheny, 2021)

Note that **grplasso** isn't computationally competitive (no majorization)

Other packages all use majorization—but with practical differences!

Scope of Packages: Loss Functions

$L(\beta \mathbf{D})$	gglasso	grplasso	grpnet	grpreg
Gaussian	✓	✓	✓	✓
Binomial	✓	✓	✓	✓
Poisson		✓	✓	✓
Multinomial			✓	
Negative Binomial			✓	
Gamma			✓	
Inverse Gaussian			✓	
Cox Proportional Hazard				✓
Squared SVM	✓			
Huberized SVM	✓			

Table 3: Loss functions available in each package.

Scope of Packages: Usability Features

Features	gglasso	grplasso	grpnet	grpreg
α tuning			✓	
Parallelization			✓	
Default method	✓		✓	✓
Formula method		✓	✓	
Orthogonalization		✓		✓
Standardization			✓	
Additive splines			✓	✓
Tensor product splines			✓	

Table 4: Usability features of each package.

Simulation Design and Analyses

Fully crossed simulation design with three factors:

1. response distribution (3 levels: Gaussian, Binomial, Poisson)
2. sample size (2 levels: $n \in \{100, 1000\}$)
3. number of predictors (10 levels: $p = 10 \times 2^m$ for $m \in \{1, \dots, 10\}$)

Data generation and analysis details:

- group size was fixed at $p_k = 5$ for all $k = 1, \dots, K$
- elements of $\mathbf{X}$ were i.i.d. $N(0, 1)$
- compared **grpnet**, **grpreg**, and **gglasso**
- calculate time to compute regularization path using 100 λ s
- repeat entire procedure 10 times for each simulation cell

Simulation Results

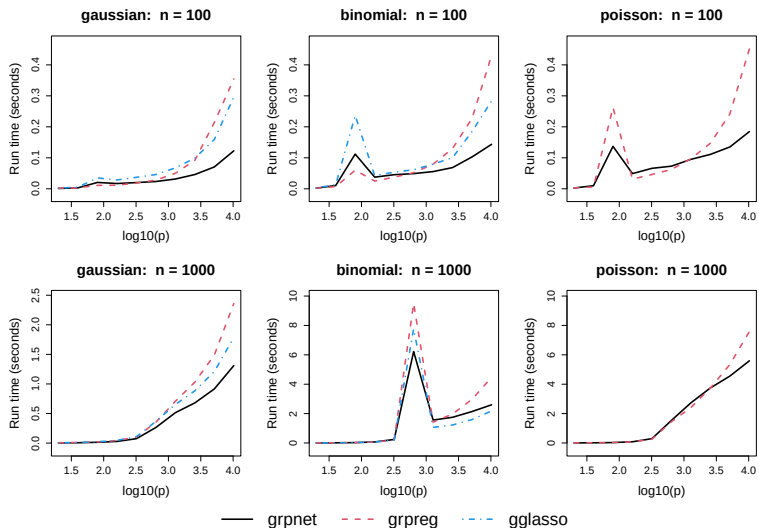


Figure 1: Amount of time to compute the regularization path.

Real Dataset Details

I consider five different datasets from the UCI-MLR (Kelly et al., 2023)

Table 5: Features of the real datasets used in the examples.

Dataset	Response	Family	n	p	K
Wine Quality	quality	Gaussian	4898	55	11
Wisconsin Breast Cancer (Original)	Class	Binomial	683	80	9
Student Performance (Math)	failures	Poisson	395	89	32
Splice-Junction Gene Sequences	border	Multinomial	3190	227	60
Gene Expression Cancer RNA-Seq	labels	Multinomial	801	101320	20264

Note: K is the number of predictors (features), whereas $p \geq K$ is the number of columns of the design matrix after expanding:

- categorical predictors using dummy coding
- continuous predictors using B-spline basis

Real Data Analyses

The `bs()` function in **splines** package (R Core Team, 2023) used to expand each continuous predictor using a B-spline basis with `df = 5`.

For the first three examples, I compared **grpnet**, **grpreg**, & **gglasso** by

- randomly splitting data into training (80%) and testing (20%) sets
- applying 10-fold CV to tune λ using the training data
- evaluating the trained models' predictions using the testing data

For the two multinomial examples, I demonstrate the functionality of the `cv.grpnet()` function and corresponding S3 plotting method.

Real Data Results: Methods Comparison

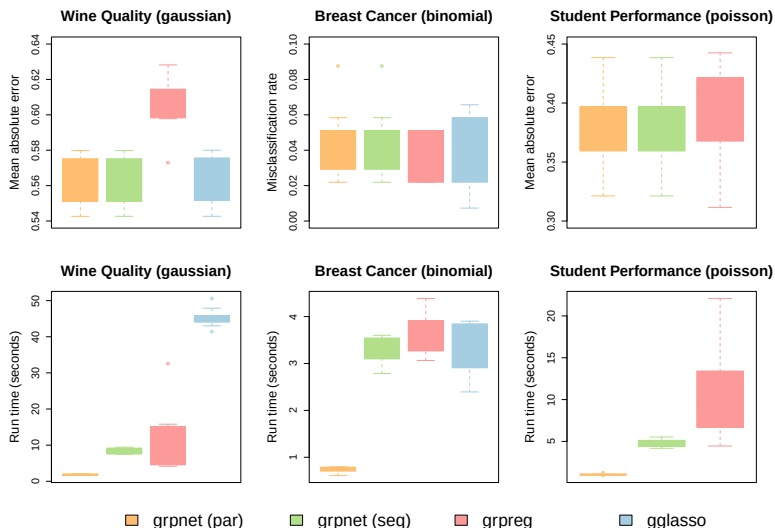


Figure 2: Prediction error for test data (top) and tuning time for training data (bottom).

Real Data Results: CV Error Curves

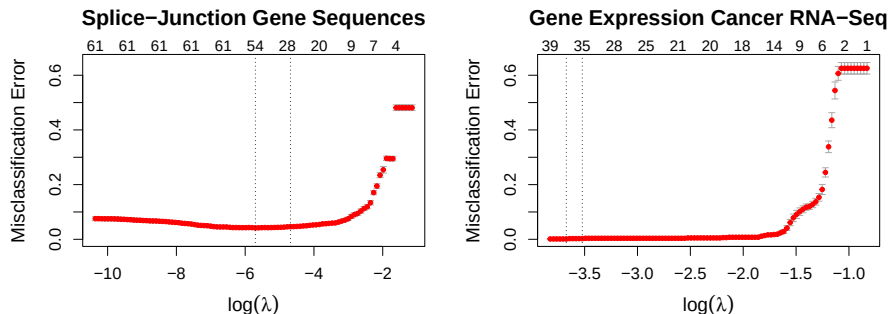


Figure 3: Cross-validation error curves (i.e., average misclassification rate by $\log(\lambda)$) with standard error bars for the multinomial examples. Within each panel, the vertical dotted lines show: the λ that minimizes the cross-validation error (left), and the largest λ that is within one standard error of the minimum (right). The numbers on the top axis show the number of non-zero groups as a function of $\log(\lambda)$.

Cancer-RNA Solution Print Out

Printing the object output from the `cv.grpnet()` function produces:

```
> cvmod.cancer
```

```
Call:  cv.grpnet.default(x = xbs, y = y, group = g, alpha = 1, parallel = TRUE,
  cluster = cl, family = "multinomial")
```

```
Measure:  Misclassification Error
```

	Alpha	Lambda	Index	Measure	SE	nzGroup	nzCoef	Df
min	1	0.02538	95	0.00125	0.001250	39	955	428.5
1se	1	0.02953	90	0.00250	0.001667	35	855	380.0

Note: the optimal solution only has 39 groups (= 38 genes + intercept)

Cancer-RNA Effect Size Plots

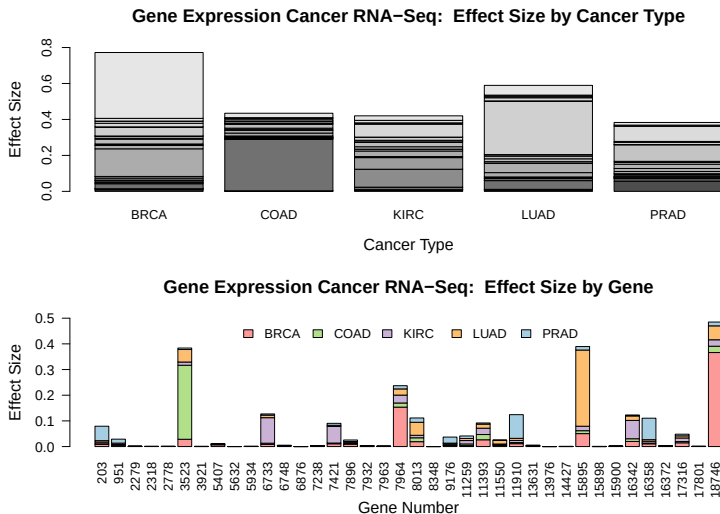


Figure 4: Effect sizes, i.e., $\xi_{kl} = n^{-1} \|C_n \mathbf{X}_k \boldsymbol{\beta}_{kl}\|^2$, for the $K = 38$ non-zero genes.

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Penalized Least Squares Functional

Consider a multiple nonparametric regression model of the form

$$Y = f(X_1, \dots, X_p) + \epsilon \quad (13)$$

where ϵ is an error term satisfying $E(\epsilon) = 0$ and $E(\epsilon^2) = \sigma^2$

Function f is estimated via penalized least-squares (PLS)

$$\frac{1}{n} \sum_{i=1}^n (y_i - f(\mathbf{x}_i))^2 + \lambda P(f) \quad (14)$$

where P is a quadratic penalty functional that quantifies the complexity (i.e., roughness) of the function f

Marginal Reproducing Kernel Hilbert Space

$\mathcal{H}_j$ denotes a RKHS on domain $\mathcal{X}_j$ with inner product $\langle \cdot, \cdot \rangle_j$ for all j .

- $\|f\|_j = \sqrt{\langle f, f \rangle_j}$, which is equal to zero if and only if $f = 0$
- $R_j(\cdot, \cdot)$ is RK: $f(x) = \langle R_j(x, z), f(z) \rangle_j$ for all $f \in \mathcal{H}_j$ and $x, z \in \mathcal{X}_j$

Suppose each marginal RKHS can be decomposed as $\mathcal{H}_j = \mathcal{H}_j^0 \oplus \mathcal{H}_j^1$

- $\mathcal{H}_j^0 = \{f : P(f) = 0, \forall x \in \mathcal{X}_j\}$ is the null space of P
- $\mathcal{H}_j^1 = \{f : P(f) > 0, \forall x \in \mathcal{X}_j\}$ is the contrast space of P

The inner-product and RK can be correspondingly decomposed:

- Inner product: $\langle \cdot, \cdot \rangle_j = \langle \cdot, \cdot \rangle_j^0 + \langle \cdot, \cdot \rangle_j^1$
- Reproducing kernel: $R_j(x, z) = R_j^0(x, z) + R_j^1(x, z)$

Tensor Product Reproducing Kernel Hilbert Space

Consider the construction of a tensor product RKHS of the form

$$\begin{aligned}\mathcal{H} &= \mathcal{H}_1 \otimes \cdots \otimes \mathcal{H}_p \\ &= \mathcal{H}_0 \oplus \underbrace{\left(\sum_{j=1}^p \mathcal{H}_j^* \right)}_{\mathcal{H}_{\{1\}}} \oplus \underbrace{\left(\sum_{j < k} \mathcal{H}_j^* \otimes \mathcal{H}_k^* \right)}_{\mathcal{H}_{\{2\}}} \oplus \underbrace{\left(\sum_{j < k < \ell} \mathcal{H}_j^* \otimes \mathcal{H}_k^* \otimes \mathcal{H}_\ell^* \right)}_{\mathcal{H}_{\{3\}}} \oplus \cdots\end{aligned}$$

defined on the product domain $\mathcal{X} = \mathcal{X}_1 \times \cdots \times \mathcal{X}_p$, where

- $\mathcal{H}_0 = \{f : f(x) \propto 1, \forall x \in \mathcal{X}_j\}$ is a space of constant functions
- $\mathcal{H}_j^* = \mathcal{H}_j \ominus \mathcal{H}_0$ is the non-constant portion of $\mathcal{H}_j$
- $\mathcal{H}_{\{1\}}$ captures the p main effects
- $\mathcal{H}_{\{2\}}$ captures the $\binom{p}{2}$ two-way interaction effects
- $\mathcal{H}_{\{3\}}$ captures the $\binom{p}{3}$ three-way interaction effects

Tensor Sum Decomposition

Given an assumed model form, we have that

$$\mathcal{H} = \mathcal{H}_0^* \oplus \mathcal{H}_1^* \oplus \cdots \oplus \mathcal{H}_K^*$$

where

- $\mathcal{H}_0^* = \{f : P(f) = 0, \forall x \in \mathcal{X}\}$ is the tensor product null space
- $\mathcal{H}_k^* = \{f : P(f) > 0, \forall x \in \mathcal{X}\}$ is the k -th contrast subspace

The corresponding inner-product and RK have the form

$$\langle \cdot, \cdot \rangle = \sum_{k=0}^K \omega_k^{-1} \langle \cdot, \cdot \rangle_k^* \quad \text{and} \quad R(\cdot, \cdot) = \sum_{k=0}^K \omega_k R_k^*(\cdot, \cdot)$$

where $\omega_k > 0$ are tuning parameters for $k = 1, \dots, K$ (with $\omega_0 = 1$).

Kimeldorf & Wahba (1971) Representer Theorem

Consider minimizing Equation (14) assuming that $f \in \mathcal{H}$ and

- $\{N_j\}_{j=1}^m$ spans the null space of the penalty (m is its dimension)
- $R_1(\cdot, \cdot) = \sum_{k=1}^K \omega_k R_k^*(\cdot, \cdot)$ is the RK for the contrast space
- $P(f) = \sum_{k=1}^K \omega_k^{-1} \langle f, f \rangle_k^*$ is the penalty functional

Theorem: Given λ and $\boldsymbol{\omega} = (\omega_1, \dots, \omega_K)$, the f_λ^* that minimizes the penalized least squares functional has the form

$$f_\lambda^*(\mathbf{x}) = \sum_{j=0}^{m-1} \beta_j N_j(\mathbf{x}) + \sum_{i=1}^n \alpha_i R_1(\mathbf{x}, \mathbf{x}_i)$$

for all $\mathbf{x} \in \mathcal{X}$, where $\boldsymbol{\beta} = (\beta_0, \dots, \beta_{m-1})^\top$ are unpenalized coefficients, and $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_n)^\top$ are penalized coefficients.

Improved Tensor Product Representer Theorem

Classic representer theorem is restrictive for product domains.

- Optimal representation depends on ω
- Works well if ω is properly tuned

Theorem: Given λ and $\{R_k^*\}_{k=1}^K$, the f_λ^* that minimizes the penalized least squares functional has the form

$$f_\lambda^*(\mathbf{x}) = \sum_{j=0}^{m-1} \beta_j N_j(\mathbf{x}) + \sum_{i=1}^n \sum_{k=1}^K \alpha_{ik} R_k^*(\mathbf{x}, \mathbf{x}_i)$$

for all $\mathbf{x} \in \mathcal{X}$, where $\boldsymbol{\beta} = (\beta_0, \dots, \beta_{m-1})^\top$ are unpenalized coefficients, and $\boldsymbol{\alpha}_k = (\alpha_{1k}, \dots, \alpha_{nk})^\top$ are penalized coefficients for k -th effect.

Comparison of Representer Theorems

The two approaches use different contrast space representations:

- classic: $\sum_{i=1}^n \sum_{k=1}^K \alpha_i \omega_k R_k^*(\mathbf{x}, \mathbf{x}_i)$
- updated: $\sum_{i=1}^n \sum_{k=1}^K \alpha_{ik} R_k^*(\mathbf{x}, \mathbf{x}_i)$

Note: “classic” is special case of “updated” with $\alpha_{ik} = \alpha_i \omega_k$ for all i, k .

Updated theorem enables more flexible tensor product smoothing:

- More flexible due to expanded tensor product RKHS
- Easier computation for estimation and tuning
- More control over knot placement within each subspace

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Component Selection and Smoothing Operator

The COSSO (Lin & Zhang, 2006) replaces the standard penalty with

$$P(f) = \sum_{k=1}^K \omega_k^{-1} \|f\|_k^*$$

where $\|f\|_k^* = \sqrt{\langle f, f \rangle_k^*}$ is the norm of the projection of f into $\mathcal{H}_k^*$.

Using the classic representer theorem, the solution minimizes

$$\frac{1}{n} \sum_{i=1}^n (y_i - f(\mathbf{x}_i))^2 + \lambda_1 \sum_{k=1}^K \omega_k^{-1} \langle f, f \rangle_k^* + \lambda_2 \sum_{k=1}^K \omega_k$$

subject to $\omega_k \geq 0$ for all $k = 1, \dots, K$.

Relation Between COSSO and Group Lasso

Suppose that each effect's representation and penalty have the form

- Representation: $\mathbf{f}_k = \mathbf{X}_k \boldsymbol{\beta}_k$
- Penalty: $\langle f, f \rangle_k^* = \boldsymbol{\beta}_k^\top \mathbf{P}_k \boldsymbol{\beta}_k$ where $\mathbf{P}_k$ is positive semidefinite

Consider reparameterizing the representation and penalty such as

$$\begin{aligned}\mathbf{f}_k &= \tilde{\mathbf{X}}_k \tilde{\boldsymbol{\beta}}_k \\ \langle f, f \rangle_k^* &= \|\tilde{\boldsymbol{\beta}}_k\|^2\end{aligned}$$

where

- $\tilde{\mathbf{X}}_k = \mathbf{X}_k \mathbf{V}_k \mathbf{E}_k^{-1/2}$ and $\tilde{\boldsymbol{\beta}}_k = \mathbf{E}_k^{1/2} \mathbf{V}_k^\top \boldsymbol{\beta}_k$
- $\mathbf{P}_k = \mathbf{V}_k \mathbf{E}_k \mathbf{V}_k^\top$ is the eigenvalue decomposition

Using updated RT (and reparam), COSSO is a group lasso problem.

More Versatile COSSO via Group Selection

Using updated RT (and reparam), consider the problem of minimizing

$$\frac{1}{n} \sum_{i=1}^n (y_i - f(\mathbf{x}_i))^2 + \sum_{k=1}^K P_1(\|f\|_k^*; \lambda_1^k) + \frac{1}{2} \sum_{k=1}^K P_2(\|f\|_k^*; \lambda_2^k)$$

where

- $P_1(\|f\|_k^*; \lambda_1^k)$ is a selection penalty (L1)
- $P_2(\|f\|_k^*; \lambda_2^k) = \lambda_2^k \langle f, f \rangle_k^*$ is a smoothing penalty (L2)

Using group lasso penalty for P_1 is equivalent to defining

$$P_1(\xi; \lambda) = \lambda \xi$$

which produces a revamped COSSO when $\lambda_2^k = 0$ for all k .

Some Improved Selection Penalties: SCAD and MCP

Could define P_1 using a group SCAD penalty (Fan & Li, 2001), such as

$$P_1(\xi; \lambda) = \begin{cases} \lambda\xi & \text{if } \xi \leq \lambda \\ -\frac{\xi^2 - 2a\lambda\xi + \lambda^2}{2(a-1)} & \text{if } \lambda < \xi \leq a\lambda \\ \frac{(a+1)\lambda^2}{2} & \text{if } \xi > a\lambda \end{cases}$$

or could use a group MCP penalty (Zhang, 2010), such as

$$P_1(\xi; \lambda) = \begin{cases} \lambda\xi - \frac{1}{2a}\xi^2 & \text{if } \xi \leq a\lambda \\ \frac{1}{2}a\lambda^2 & \text{if } \xi > a\lambda \end{cases}$$

where $a > 2$ (SCAD) or $a > 1$ (MCP) is a tuning/hyper-parameter.

Simulated Example: Data Generating Model

Data Generating Mean Function Form:

$$\mu(X) = \alpha_1 f_1(X_1) + \alpha_2 f_2(X_2) + \alpha_3 f_3(X_1, X_2)$$

where

- f_1 and f_2 are the main effect functions
- f_3 is the interaction effect function
- $(\alpha_1, \alpha_2, \alpha_3)$ are varied to control size of effects
- 3 active terms: X_1 , X_2 , and $X_1 : X_2$

Independently generate data of the form

$$Y_i = \mu(X_i) + \epsilon_i$$

where X_{ij} are i.i.d. $U[0, 1]$ and $\epsilon_i \sim N(0, \sigma^2)$.

Simulated Example: Component Functions

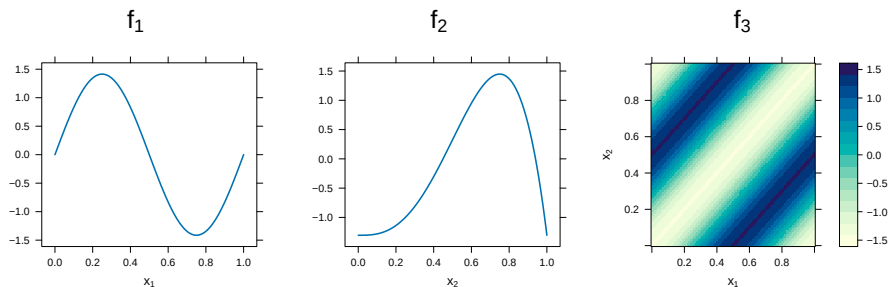


Figure 5: Component functions used in simulation. $f_1(x) \propto \sin(2\pi x)$ (sine function), $f_2(x) \propto \text{Beta}(x; 4, 2) - 1$ (beta density function), and $f_3(x_1, x_2) \propto -\cos(2\pi|x_1 - x_2|)$ (cosine function). All component functions were centered (to have mean zero) and scaled (to have unit variance) for the domain $\mathcal{X} = [0, 1] \times [0, 1]$.

Simulated Example: Data Generating Functions

Consider three different mean functions:

(A) $\alpha_1 = \alpha_2 = \alpha_3 = 1$ (equal strength effects)

(B) $\alpha_1 = \alpha_2 = 1.2$ and $\alpha_3 = 0.6$ (weak interaction effect)

(C) $\alpha_1 = \alpha_2 = 0.6$ and $\alpha_3 = 1.8$ (weak main effects)

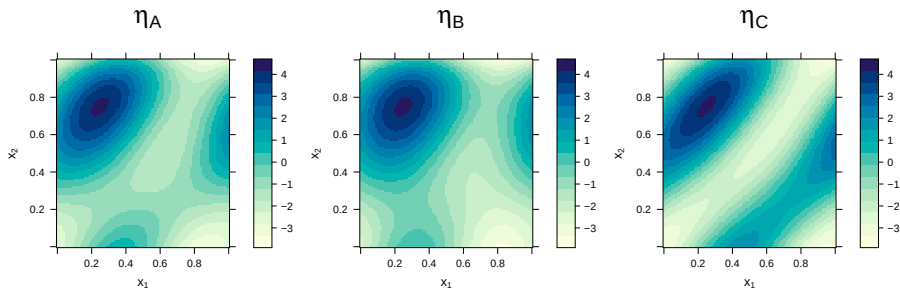


Figure 6: Mean functions formed via linear combinations of the component functions.

Simulated Example: Design and Analyses

Fully crossed design that manipulates two factors:

- $n \in \{100, 250, 500, 750, 1000\}$
- $p \in \{10, 20\}$

Cubic spline basis with 5 df for each marginal

- Each main effect has 5 coefficients
- Each two-way interaction effect has 25 coefficients

Fit model with all p main effects and all $\binom{p}{2}$ two-way interactions:

p	$\binom{p}{2}$	K	# coefs
10	45	55	1175
20	190	210	4850

Simulated Example: Model Recovery Results ($\sigma = 1/2$)

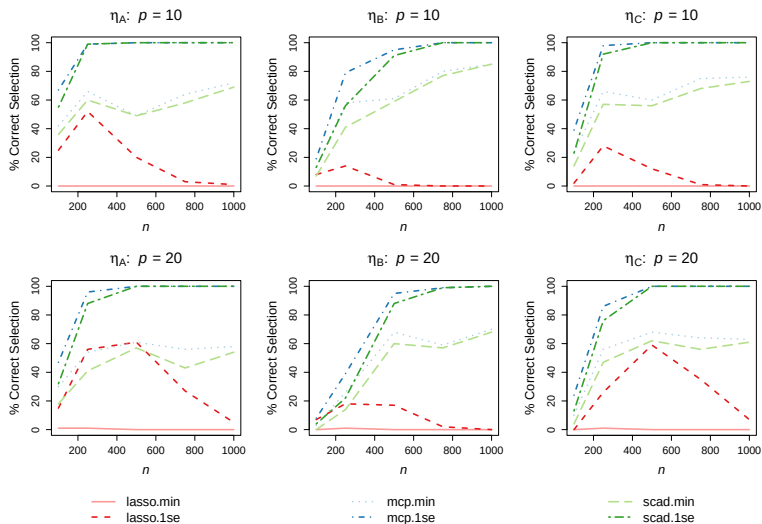


Figure 7: Percentage of time 10-fold CV selected the correct model when $\sigma = 1/2$.

Simulated Example: Model Recovery Results ($\sigma = 1$)

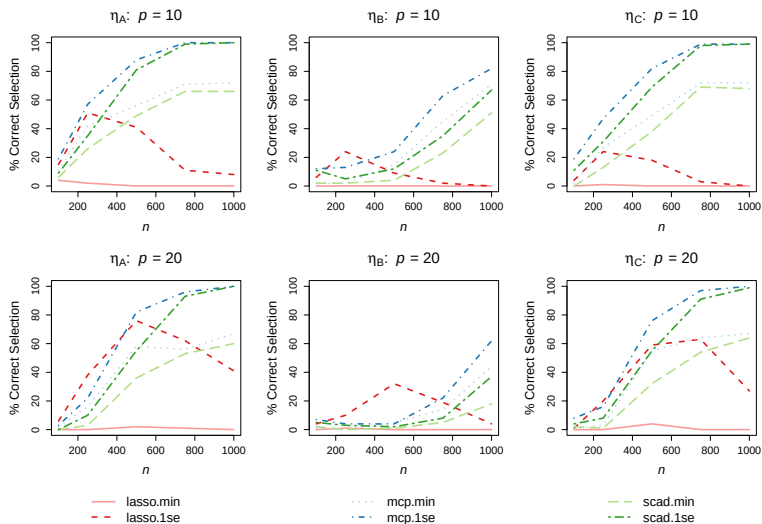


Figure 8: Percentage of time 10-fold CV selected the correct model when $\sigma = 1$.

Real Data Example: Dataset and Analyses*

Data from Adolescent Brain Cognitive Development (ABCD) study

- $n = 7979$ children ages 9-10 years old
- 21 collection sites across the USA (including UMN)

Response: Attention Problems subscale of Child Behavior Checklist

Predictors:

- 78 pairs of resting state network correlations (13 brain networks)
- Parent history of alcohol and drug use
- Collection site ID, Handedness, Income, and Education
- Demographic information (age, sex, race)

*Joint work with Kelly Duffy (QPM)

Real Data Example: Network Connectivity Importance

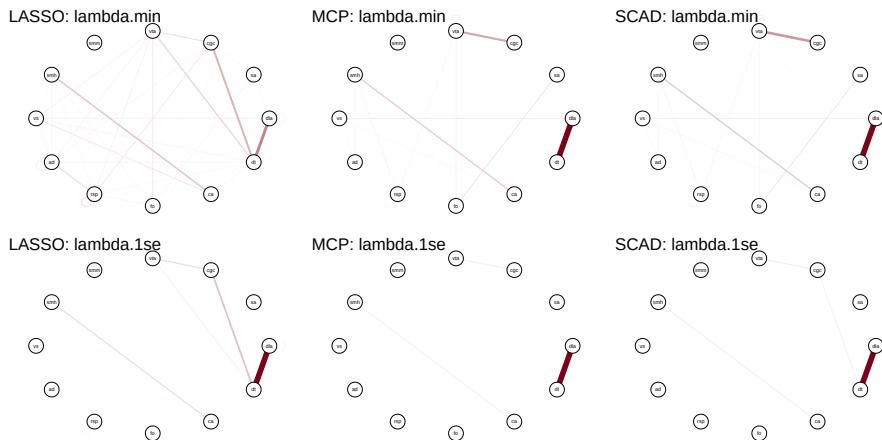


Figure 9: Importance index for each pairwise network correlation using different penalties and model selection rules. Results are averaged across 10 different random splits.

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Take Home Points

Model selection and smoothing is possible but...

- Group MCP $>$ group SCAD (group LASSO doesn't work)
- Need a large enough sample size to find active effects
- Finding weak interaction effects needs large n (or large SNR)

Interestingly, the **1se** rule outperformed the **min** rule (by a lot).

- The **1se** rule always improved as n increased
- The **min** rule was always worse than the **1se** rule
- Maybe the **1se** rule should be the default?

Room for Improvement

Hyperparameter Tuning

- ω_k could be tuned or adaptively estimated
- a parameter (for SCAD and MCP) could be tuned

Knot Placement

- For additive models, univariate quantiles work well as knots
- For interactions, better strategy needed (multivariate quantiles?)

Statistical Inference

- Bootstrap can be used, but computationally costly
- Bayesian interpretation of penalty may be useful

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